

Lightweight and Longitudinal Constraint-Based IP Geolocation via Multi-Stage Landmark Selection

Anonymous Author(s)

Abstract

Active IP geolocation produces measurement-based, auditable location estimates—a property that proprietary databases cannot offer—but at significant operational cost. Each geolocation campaign orchestrates hundreds of landmarks—vantage points at known geographic locations—to probe a target, with an end-to-end latency on the order of minutes per target. Reliable use further requires periodic re-verification, since IP-to-location bindings drift over time as prefixes are reassigned, compounding the operational cost.

We extend CBG++ to enable longitudinal IP geolocation at scale. We propose a three-stage landmark selection algorithm that substantially reduces per-target probing costs. The design is informed by an empirical analysis showing that a single fast-RTT landmark deterministically achieves city-level accuracy under sufficient proximity, while multi-landmark intersection partially recovers boundary cases. Through cross-validation on 795 RIPE Atlas anchors, fast-RTT landmarks carry the essential geolocation signal but are unavailable for 22% of targets due to coverage sparsity—directly motivating our multi-stage fallback strategy.

Our pipeline achieves 70.3% city-level accuracy (≤ 40 km), 4.8 percentage points above an exhaustive full-landmark baseline, while issuing roughly $10\times$ fewer measurements per target (~ 80 vs. ~ 788).

1 Introduction

IP geolocation maps an Internet host to its physical location and underpins a broad spectrum of applications. These range from operational network diagnostics—such as route prediction [9], path troubleshooting [13], and censorship analysis [11, 26]—to security and commercial services, including botnet detection [10], proxy verification [34], and fraud prevention [4]. Passive IP-to-location databases combine WHOIS records, regional registry allocations, and routable information with proprietary inference signals to provide millisecond-scale lookups [24]. Their methodology is rarely disclosed [28], and their stability degrades over time as IP prefixes are reassigned [17]. Active geolocation instead infers a target’s position from network measurements. In Constraint-Based Geolocation (CBG) [18], the round-trip time (RTT) from each *landmark* host—a server of known location—to the target is converted into an upper bound on geodesic distance via the speed of light, and per-landmark constraint disks are intersected to localize the target.

Active geolocation, however, requires substantial measurement infrastructure. Each query requires probing across many landmarks plus periodic calibration of the latency-distance mapping, both translating into ongoing measurement overhead; accuracy further depends on landmark proximity to the target. Targets’ physical locations are also not static—cloud-hosted services migrate between datacenters, and residential ISPs periodically reassign prefixes [27]—so reliable geolocation requires *periodic* re-verification, with cost compounding as the same target is re-checked over time. Existing active-geolocation techniques have not been engineered for this regime; they optimize the one-shot accuracy of a single campaign rather than the amortized cost of repeated re-verification.

Prior work has pursued two complementary goals: improving localization accuracy [23, 34, 35] and reducing measurement cost. Cost-reduction efforts share a common premise of *landmark proximity*—that a sufficiently close landmark exists for the target. Hu et al. [20] probe representative IPs in each target’s /24 prefix to select the ten nearest vantage points; TinyG [22] sequentially picks probers expected to yield very low RTTs; RIPE IPMap [14] infers location from the closest available probe; and GeoResolver [29] uses DNS redirection patterns to identify likely-close landmarks before measurement. In regions with sparse landmark coverage, the proximity premise breaks down, leading to a sharp cost–accuracy trade-off. An orthogonal thread leverages operator-disclosed location data (*geofeeds* [5]), but adoption remains low and even well-formed feeds exhibit non-trivial inaccuracies that prevent them from serving as a reliable standalone solution [25]. Despite these advancements, the complementary problem of amortizing measurement costs across repeated re-verifications remains largely unaddressed.

Our work fills this longitudinal-cost gap while mitigating the reliance on dense landmark proximity. We extend CBG++ [34] to support periodic re-verification of IP geolocation targets at low amortized cost at scale. Following prior work [12, 15, 22], we target city-level accuracy (≤ 40 km centroid error), the threshold sufficient for most location-based services. We propose a three-stage landmark selection algorithm informed by an empirical analysis of where CBG-style geolocation succeeds, where multi-landmark intersection helps recover boundary cases, and where the current landmark mesh is fundamentally insufficient. We complement this with mechanisms for filtering inherently unstable IPs

(anycast detection), validating the framework on IPv6 anchors, analyzing calibration parameter stability over time, and a lightweight re-verification probe that triggers full re-localization only when landmark RTTs shift beyond a threshold.

This paper makes the following contributions:

- An empirical decomposition (via cross-validation on 795 RIPE Atlas anchors) that identifies three distinct geolocation regimes: deterministic single-landmark sufficiency, multi-landmark recovery, and structural coverage limits.
- A three-stage landmark selection algorithm that substantially reduces per-target probing cost while preserving city-level accuracy where achievable.
- Anycast and CDN pre-filtering, integrating existing detection techniques into the longitudinal geolocation pipeline.
- An IPv6 cross-validation demonstrating that the approach generalizes across address families.
- A 30-day stability analysis of calibration parameters, paired with a comparison of LP-based and quantile regression calibration.

The remainder of the paper is organized as follows. Section 2 reviews CBG++ and the role of landmarks. Section 3 presents our geolocation framework—the calibration trade-off between tightness and coverage, anycast and CDN pre-filtering, the CBG precision score, the consistency check, and the re-verification mechanism. Section 4 develops the three-stage landmark selection algorithm. Section 5 reports our evaluation, covering IPv4 and IPv6 cross-validation, the calibration comparison, anycast/CDN filtering performance, coverage-stratified analysis, selection performance, longitudinal operation, and an end-to-end performance comparison. The paper concludes with a discussion of related work (Section 6), limitations (Section 7), ethical considerations (Section 8), and conclusion in Section 9.

2 Background

2.1 Constraint-Based Geolocation: CBG++

CBG++ builds on Constraint-Based Geolocation (CBG) [18], an active geolocation method that infers a host’s location by intersecting distance-based constraints derived from round-trip time (RTT) measurements to known landmarks. While CBG is effective, it can underestimate the distance a packet may travel, resulting in constraint regions that are too small to contain the true location. This underestimation often arises from calibration inaccuracies caused by transient network conditions [34]. CBG++ mitigates this by introducing physical plausibility bounds during calibration and multilateration.

2.1.1 Calibration. The calibration phase establishes a landmark specific *bestline*, a linear model mapping RTT to geographic distance. The *bestline* is fitted to be as close as possible to landmark-to-landmark observations while remaining strictly below all data points to approximate maximum propagation speeds. To ensure physical plausibility, CBG++ constrains the *bestline* slope between a theoretical *baseline* of 200km/ms (two-thirds of c , signal propagation in fiber) and a *slowline* of 84.5km/ms. The slowline represents the minimum speed required to traverse half the Earth’s circumference with the 237ms threshold typically associated with geostationary satellite hops. In addition, the algorithm utilizes the minimum observed RTT per distance bin to filter out queuing delays and more accurately reflect geographic distance.

2.1.2 Multilateration. The final location is derived through a two-stage iterative refinement process. Disks whose radius is too large to meaningfully constrain the target are discarded before intersection. First, the algorithm identifies the largest subset of *baseline* disks (calculated at 200km/ms) that share a non-empty intersection, establishing a conservative *baseline* region. Second, *bestline* disks that do not overlap this region are discarded as outliers. The algorithm then identifies the largest subset of the remaining *bestline* disks with a non-empty intersection to produce the final prediction area. This hierarchical strategy ensures the resulting region is both geographically precise and resilient to the underestimation errors inherent in standard CBG.

2.2 Landmarks

Our framework utilizes RIPE Atlas anchors as primary landmarks due to their high operational stability and verified geographic coordinates. Anchors are server-grade hosts that continuously perform mutual measurements, uploading a constant mesh of anchor-to-anchor RTTs to a public database. This continuously updated mesh provides recent measurements that we use to calibrate the latency-distance mapping (§3). Anchors are institutionally hosted, well-connected, and continuously available. Although the fleet evolves over time as nodes are commissioned and decommissioned, each active anchor remains a reliable reference point—a property our framework relies on for calibration, landmark selection, and the Speed-of-Light (SoL) violation checks used in anycast pre-filtering.

We leverage probes as a secondary tier of landmarks, invoked when the anchor pool offers insufficient spatial coverage relative to a target. Unlike institutionally-hosted anchors, RIPE Atlas probes are small, low-cost devices deployed across a far denser and more diverse range of residential and enterprise networks, providing fine-grained geographic granularity. Notably, while anchors are heavily concentrated in

Table 1: Performance of CBG++ (LP) across 788 RIPE Atlas anchors (Jan 22, 2026).

Metric	CBG++ (LP)
<i>Geographical accuracy</i>	
In-region	77.0%
Median centroid error	12.6 km
Mean centroid error	145.1 km
Within 25 km	60.8%
Median edge distance	25.8 km
<i>Jurisdictional validity</i>	
Country match	98.1%
<i>Spatial precision</i>	
Median area	3,343 km ²
Mean area	1,086,857 km ²
Median # countries	1
Mean # countries	3.5

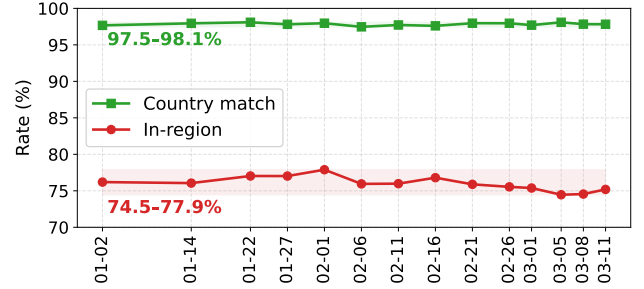
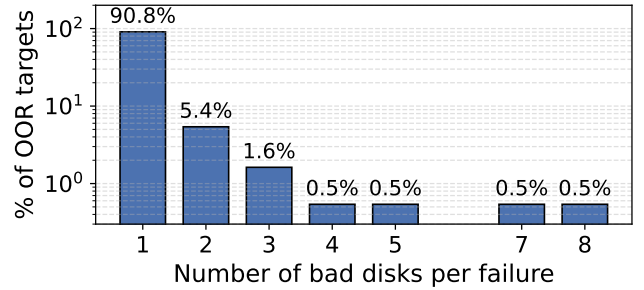
Europe and North America, probes extend coverage to under-represented regions such as Africa and South America. By selectively integrating these non-anchor nodes—filtered for stability and location accuracy—we obtain a more angularly diverse set of constraints to better ‘bracket’ each target.

3 Our Framework: Geolocation Engine

3.1 Initial Analysis & Motivation

We evaluate CBG++ [34] through cross-validation on 795 globally distributed RIPE Atlas anchors. We analyze the resulting uncertainty polygons across three complementary categories. First, for *geographical accuracy*, we follow established practice in constraint-based geolocation [14, 23, 35] by measuring whether the truth falls inside the region (in-region rate), the geodesic error from centroid to truth, and the distance from truth to the region boundary. Second, for *jurisdictional validity*, we report country-match rate. Third, for *spatial precision*, we report region area and the number of intersected countries. These metrics capture two distinct dimensions of CBG++ output—a point estimate (centroid) and a bounding region whose size reflects how tightly the available constraints localize the target. Centroid accuracy at the city level (≤ 40 km) is our primary metric, while the region-level properties serve as diagnostics that identify where and why specific predictions succeed or fail.

While CBG++ achieves a 98.1% country match, its in-region containment (coverage of true coordinates) is limited to 77.0% (Table 1). To determine whether this gap reflects a transient artifact or a structural limitation, we conducted a longitudinal study across 14 dates over 10 weeks on RIPE Atlas anchors. The results (Figure 1) confirm that the gap is structural: in-region containment consistently remains


Figure 1: Longitudinal performance of CBG++ (LP) over 10 weeks (Jan–Mar 2026).

Figure 2: Distribution of incorrect disks per out-of-region (OOR) failure (n=185, Jan 22, 2026).

within the 74.5–77.9% range, while country-level matching stays stable at $\approx 98\%$. To localize the source of this gap, we decompose CBG++ into its two stages: individual disk construction and disk intersection. We find that 99.72% of *bestline* disks that enter the intersection contain the true target, indicating that calibration is not the source of error. Instead, the loss occurs during intersection: only 77% of final regions contain the truth, despite nearly every input disk being correct. The gap is therefore combinatorial rather than calibration-driven. A small number of incorrect disks corrupts the intersection.

We further investigate CBG++’s DFS intersection algorithm to identify the specific disks responsible for failures.¹ Among all 185 out-of-region (OOR) targets, the DFS-selected subset contains at least one incorrect disk in every case. Notably, 90.8% of these failures involve exactly one such disk (Figure 2). These failures are near-misses: the median distance from the true location to the region boundary is 25.8 km (Table 1). This raises the question of whether the in-region failures are catastrophic or merely region-shape artifacts—a question we revisit in §3.3.

¹Re-run as a separate instrumented pass; count differs from the CBG++ (LP) column in Table 1 by 4 targets due to independent timeout handling.

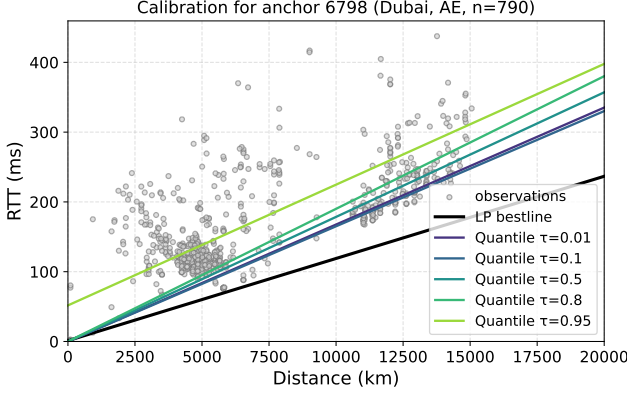


Figure 3: Calibration analysis for anchor 6798, showing the LP baseline and various quantile regression fits (Jan 22, 2026).

3.2 Calibration: LP Bestline

CBG++ converts measured RTTs into distance upper bounds using a calibrated RTT–distance line, termed the *bestline*. We adopt the linear-programming (LP) formulation from CBG++ [34], which fits the line strictly below all calibration points to satisfy the feasibility constraint. Figure 3 illustrates this fit alongside several quantile-regression (QR) alternatives.

Quantile regression with high τ produces tighter individual disks by permitting a parameterized fraction of observations to fall below the calibration line. However, as we detail in §5.2, our cross-validation sweep reveals a structural failure mode: these restrictive QR disks frequently fail to mutually overlap, forcing the intersection algorithm to fall back to less-informative subsets. This inconsistency often causes the final polygon to collapse to the physical-limit intersection, paradoxically inflating the area by 2–7 $\times$ while providing no accuracy gain. In contrast, LP’s per-bin minimum preprocessing extracts a more mutually-consistent lower envelope, yielding both smaller polygons and superior accuracy in regions with sparse infrastructure. We therefore retain LP as our calibration baseline.

While a static buffer on the calibration line may seem appealing, our preliminary tests indicate that it does not resolve the underlying issue. Calibration error varies significantly per anchor, and a uniform buffer fails to account for this heterogeneity—unnecessarily expanding typical disks while still failing to cover the worst-case ones.

3.3 Intersection: Max-Subset Inclusion

We adopt the maximum-subset intersection from CBG++ [34], a depth-first search finds the largest subset of landmark disks with non-empty intersection, with smaller area as tiebreaker.

This naturally tolerates a small number of inconsistent constraints.

Motivated by prior findings that emphasize short-RTT landmarks’ importance for high-precision geolocation [12, 22], we tested two variants: *RTT-weighted maximum subset* and *trusted-core then expand* (mandatorily including the K shortest-RTT disks). To further investigate the source of the 77% containment gap, We also ran an oracle counterfactual removing disks identified (via ground truth) as not containing the target. On 794 anchors, no variant improved city-level accuracy (≤ 0.3 pp), and oracle removal made centroid error worse (+9.82 km mean Δ err for out-of-region targets; 17.2% targets worsened vs. 13.8% improved).

This reveals the precision–containment trade-off central to CBG: even disks that exclude the true location often serve as binding constraints that anchor the centroid. We find that the max-subset algorithm’s tendency to include geometrically consistent disks—even if empirically incorrect—is preferable to their removal, which releases the centroid to drift and degrades precision. Since our tested variants failed to outperform this baseline, we conclude that further improvements likely require probabilistic frameworks [23] or auxiliary data beyond RTT, both of which we leave to future work.

3.4 Anycast and CDN Pre-filtering

CBG++ assumes that each target IP corresponds to a single physical location. This assumption is increasingly violated by anycast deployments—public DNS, CDN front-ends, and DDoS-mitigation services [8, 19]—whose replicas advertise the same address from geographically dispersed sites. Such targets produce no meaningful CBG regions regardless of calibration quality. We therefore filter apparent anycast targets out before intersection, adopting the speed-of-light (SoL) violation test introduced by iGreedy [8] and recently reused as a confirmation stage in the LACeS census system [19]. We make no novelty claim for the test itself; our integration is an engineering robustness step.

For two vantage points p, q with geodesic distance $d(p, q)$ and one-way delay estimates $\delta(p, t), \delta(q, t)$ to a target t , we compute upper-bound radii

$$d(p, t) = c_f \cdot \delta(p, t),$$

with $c_f = 200$ km/ms (speed of light in fiber [18]). A violation

$$d(p, q) > d(p, t) + d(q, t)$$

implies that p and q are in contact with distinct replicas: since $d(p, t)$ and $d(q, t)$ over-estimate the true distances, no unicast target can satisfy this inequality [8].

We apply the test to three fixed pairs of geographically distant landmarks and flag a target if any pair reports a

violation, a lightweight variant of iGreedy’s full maximum-independent-set formulation [8]. Three pairs suffice to exclude globally-distributed anycast from calibration; the test misses replicas clustered within a single city [19], which is harmless for our purpose: CBG localizes them correctly as a single point.

4 Our Framework: Landmark Selection

Our problem is to choose a small subset of anchors so that CBG++ remains accurate, fast, and robust across globally distributed targets. In contrast to prior work that relies on a single low-RTT prober (e.g., TinyG [22]), we select a subset of anchors that collectively provide directionally diverse geometric constraints, trading reliance on an extremely close landmark for robustness against delay noise. Formally, let X denote the full anchor set and $Y \subseteq X$ a selected subset, and let $D(L)$ denote the CBG++ location estimate produced from landmark set L for a given target. Since we care about city-level accuracy, our goal is to choose Y such that the subset estimate closely reproduces the full-set estimate:

$$\text{maximize } \sum_{t \in T} \mathbb{1}[H(D_t(X), D_t(Y)) \leq 40 \text{ km}], \text{ s.t. } |Y| \leq k, \quad (1)$$

where T is the set of targets, H is the haversine distance, and k is the subset budget.

4.1 Stage 1: ϵ -net Anchor Selection

The simplest baseline—drawing k anchors uniformly at random from X —fails for two reasons. First, the anchor mesh is far from uniform: its median nearest-neighbour distance is only 3 km, while a uniform distribution over the same region would give a median of 364 km. A uniform sample therefore over-represents densely clustered metropolitan regions. Second, an isolated anchor contributes a loose disk that sits at the periphery of the intersection and does not help tighten the polygon. We need a sample that is geographically diverse *and* excludes structurally isolated outliers.

We use a modified ϵ -net, a sampling technique whose key guarantee is that for any geometric range, the fraction of the sample inside the range approximates the fraction of the full population within ϵ . We instantiate the metric D as pairwise minimum RTT between anchors; using RTT rather than great-circle distance keeps anchors in the same city distinguishable when their RTTs to the target diverge—two Frankfurt anchors are no longer treated as “the same point,” which prevents the sample from collapsing dense clusters.

The sample size is

$$|Y| \geq \frac{1}{\epsilon^2} (d \ln(d/\epsilon) + \ln(1/\delta)),$$

with $d = 3$, the VC-dimension of the family of geographic disks—the natural range space for CBG++, since each anchor contributes a single disk whose intersection forms the candidate polygon. The Haussler–Welzl bound then guarantees that the polygon produced by the sample approximates the polygon produced by the full anchor population, with quantifiable error. The parameter ϵ trades approximation quality against sample size: smaller ϵ yields a tighter approximation but a larger sample. When $|Y|$ exceeds the number of available anchors for a target—for instance, in a region with few anchor neighbours—we use the full set instead.

Rather than draw $|Y|$ anchors uniformly at random—which, given the clustering described above, would simply repeat the random-sampling failure—we build the net constructively via farthest-point sampling (FPS). Starting from a random seed, at each step we add the anchor whose nearest already-selected neighbour is farthest under D :

$$L_{\text{next}} = \arg \max_{L_i \notin S} \min_{L_j \in S} D(L_i, L_j).$$

FPS is a 2-approximation for the k -centre problem and is invariant to local density: a region containing a hundred near-coincident anchors contributes only a single representative to the net. This is what lets the sample “look uniform” even when the underlying distribution is heavily skewed: the ratio of European-to-African density in the sample is close to one, while it is roughly fifty in the full population.

FPS has a complementary failure mode. By always reaching for the farthest-yet-uncovered anchor, it is biased toward isolated outliers—a lone anchor in the middle of the ocean, or in a sparsely instrumented region. Such anchors contribute loose disks that sit at the periphery of the intersection and weaken rather than tighten the candidate polygon. We therefore apply an isolation filter after FPS. Each selected anchor receives a k -nearest-neighbour distance score over the full anchor set, and the most isolated fraction (typically the top 10%) is discarded. What remains is a compact, geographically and network-topologically diverse subset whose intersection polygon closely matches the one produced by the full population, while being small enough for the disk-intersection search to terminate within a bounded time budget.

4.2 Stage 2: Short-RTT Refinement

After ϵ -net selection and the proximity refinement step, the candidate polygon can still remain larger than necessary. Proximity refinement adds landmarks that are geographically close to the current polygon’s centroid, but geographic closeness is only a proxy for what we actually need: short round-trip time, which yields the tightest calibrated disk. A landmark whose great-circle distance to the centroid is large can still report a very small RTT to the target when

the two are connected by a short, direct network path; conversely, two landmarks in the same city can have very different RTTs to a target hosted abroad, depending on which transit providers each peers with. Stage 2 therefore adds a second refinement pass that selects landmarks directly by their observed RTT.

Let P denote the set of landmarks already chosen by the ϵ -net and proximity refinement, and let A denote the full set of landmarks reaching the target. We sort the remaining candidates $A \setminus P$ by ascending observed minimum RTT to the target and append the k smallest to the selection:

$$S_{\text{rtt}} = \arg \min_{L \in A \setminus P}^{(k)} \text{RTT}(L, \text{target}).$$

The intuition is direct. Each landmark L contributes a disk of radius $r(L) = \text{cal}_L(\text{RTT}(L, \text{target}))$, where cal_L is the per-landmark calibration mapping observed RTT to a distance bound. Because this map is monotone, the smallest RTTs yield the smallest radii, and the smallest-radius disks most aggressively constrain the intersection. Adding even a handful of these tight disks can collapse a polygon that proximity refinement alone had left several thousand kilometres wide.

This step is particularly important for targets where geographic and network closeness diverge. Island targets and targets in sparsely instrumented regions are the canonical examples: a target on Réunion is geographically closest to landmarks in Madagascar or Mauritius, but its smallest-RTT landmark may sit in metropolitan France, reached via a direct submarine cable. Proximity refinement, anchored on the centroid of the current polygon, would never select that landmark; RTT refinement reaches it directly, and because the resulting disk is tight despite its geographic distance, it supplies the constraint that pulls the polygon onto the correct island.

A natural question is why we do not perform RTT-based selection first and dispense with the proximity step entirely. In practice, the set of smallest-RTT landmarks tends to be drawn from a single network neighbourhood and to lie along the same general direction from the target, producing a polygon that is small in area but elongated along that direction. Proximity refinement supplies the angular diversity required for the intersection to be tight in every direction, while RTT refinement supplies the radial tightness. Used in sequence, the two complement each other: proximity refinement fixes the polygon’s angular extent, and RTT refinement collapses its radial extent.

4.3 Stage 3: Atlas Probe Fallback

Stages 1 and 2 reach city-level accuracy for most targets. For the remainder, where the available short-RTT anchors do not produce a tight enough constraint region, Stage 3 serves as a *fallback*. It augments the constraint set with up to $K = 10$

RIPE Atlas probes drawn from the broader, non-anchor probe mesh, ranked by RTT to the target.²

For each target t , the candidate pool is the set of Atlas probes that pass the same reliability filter as anchors. At least three successful pings to known anchors and a non-degenerate CBG++ LP calibration. Because the round-trip time from a candidate probe to t is unknown until measurement, our selection relies only on publicly available probe metadata. We sort candidates by ascending great-circle distance from the Stage 2 region centroid, take the top $N = 10$, and within that pool greedily select K probes that maximize the minimum pairwise angular distance (bearing from the Stage 2 centroid). The first probe is the one closest to the centroid; each subsequent probe is the unselected candidate whose bearing is furthest from those already chosen. We then ping only these K probes.

For each selected probe p , we build an empirical disk using its per-probe CBG++ LP calibration, identically to an anchor disk (§3.2). The probe disks are passed *jointly* with the Stage 2 anchor disks into our intersection routine (§3.3), which performs a depth-bounded DFS over disk subsets and returns the largest subset whose intersection is non-empty. The centroid of the resulting region is reported as Stage 3’s location estimate.

5 Evaluation

5.1 Cross-Validation (IPv4 & IPv6)

While Rimlinger et al. [29] demonstrated that DNS-redirection-based landmark selection works for IPv6 point estimates, constraint-based geolocation—which produces a polygonal feasible region rather than a point—has not been validated for IPv6 in prior work. We close this gap with a symmetric leave-one-out cross-validation on the RIPE anchor mesh (Jan 22, 2026). Each anchor in turn becomes the target while the remainder serve as landmarks, with calibrations recomputed per fold to exclude the target.

Of the full mesh, 788 anchors yielded successful IPv4 geolocations and 605 yielded IPv6. We report on the intersection set of 592 dual-stack anchors successful in both families (Table 2); only 13 anchors (1.6% of the IPv6 set) succeeded in IPv6 but failed in IPv4, so measurement availability is not a meaningful bottleneck.

The pipeline transfers to IPv6 with only negligible degradation in accuracy. City-level accuracy (≤ 40 km) drops 4.9 pp (66.7% to 61.8%), in-region rate drops 2.4 pp (76.7% to 74.3%), and country match is essentially unchanged (98.1% vs. 98.0%). Spatial precision is preserved. The median feasible region is

²The non-anchor Atlas mesh is approximately an order of magnitude denser than the anchor mesh ($\sim 12,000$ vs. ~ 900 globally), providing extra coverage in regions where short-RTT anchor availability is sparse.

Table 2: Cross-validation performance comparison between IPv4 and IPv6. Results are derived from a symmetric intersection set of 592 RIPE anchors supporting both address families (Jan 22, 2026).

	IPv4	IPv6
<i>Geographical containment</i>		
In-region rate:	76.7%	74.3%
Median error:	11.6 km	15.3 km
Mean error:	129.8 km	169.9 km
Median edge dist:	23.9 km	27.4 km
Within 25 km:	368/592 (62.2%)	334/592 (56.4%)
Within 40 km (city-level):	395/592 (66.7%)	366/592 (61.8%)
Within 100 km:	451/592 (76.2%)	426/592 (72.0%)
<i>Jurisdictional validity</i>		
Country match rate:	98.1%	98.0%
<i>Spatial precision</i>		
Median area:	2,681 km ²	2,945 km ²
Mean area:	825,112 km ²	1,409,447 km ²
Median # countries:	1	1
Mean # countries:	3.1	3.6

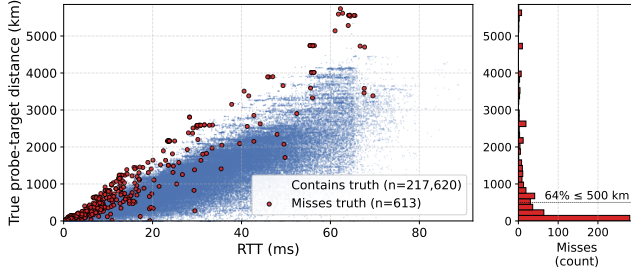


Figure 4: Per-disk containment rate for CBG++ empirical disks (LP bestline) on Jan 22, 2026.

within 10% across families (2,681 km² vs. 2,945 km²), with both medians spanning a single country.

5.2 Calibration Choice (LP vs QR)

As discussed in §3.2, we evaluate whether QR’s probabilistic relaxation can outperform LP’s hard lower bound. Across our anchor mesh, LP’s per-disk containment rate is 99.72%—near-perfect—but its 0.28% failure mode reveals a fragility we call the *Precision Paradox*: 31% of failing disks occur within 50 km of the target and 64% within 500 km (Figure 4). Failures concentrate where high resolution is most achievable; a few milliseconds of last-mile delay can displace a tight, short-range disk past the true target. This explains why containment, not headline disk-radius shrinkage, is the dimension that matters.

Quantile regression replaces LP’s hard lower bound with a soft envelope: τ permits a fraction of points to fall below

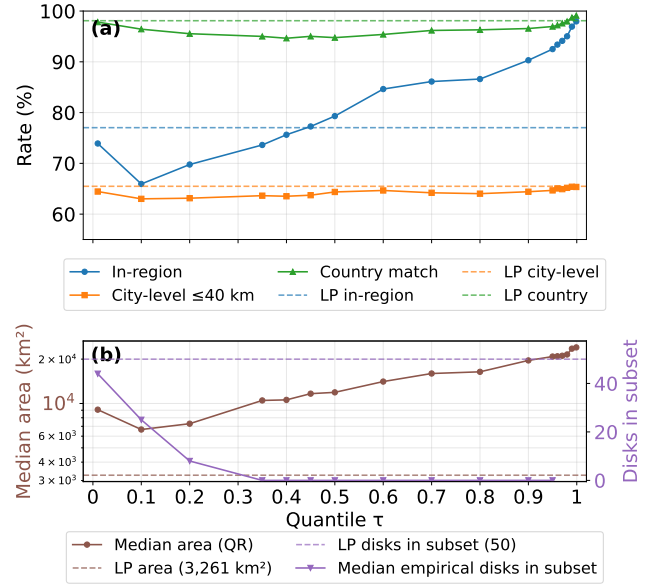


Figure 5: LP vs. quantile-regression (QR) calibration sweep on Jan 22, 2026. (a) Accuracy: QR (solid) vs. LP (dashed). In-region rate rises with τ due to polygon inflation, while city-level accuracy remains static. (b) Precision: Beyond $\tau \geq 0.35$, QR disks lose mutual overlap, causing the polygon to collapse and inflating the area 2–7× relative to LP

the calibration line. In principle this could shrink disks while preserving containment. In practice it does not. Figure 5 sweeps $\tau \in [0.01, 0.999]$ on the same mesh. As τ rises, individual disks tighten but cease to mutually overlap. The median empirical disks retained in the DFS-selected intersection collapses from 44 at $\tau=0.01$ to zero from $\tau \geq 0.35$ onward (panel b). More than half of anchors retain no empirical constraint at all, and the polygon falls back to the physical-limit intersection.

The result, in panel (a), is that QR’s in-region uplift reflects polygon-area inflation (2–7× over LP) rather than better localization. City-level accuracy (≤ 40 km) stays within 63.0–65.5% across all τ and never crosses LP’s 65.5%. We therefore retain LP as our calibration baseline.

5.3 Anycast/CDN Filtering Performance

We evaluate our speed-of-light (SoL) pre-filter across four datasets. The test correctly identified all 12 responsive DNS root servers [31] (G-root was unresponsive), matching the performance of LACeS [19], and produced zero flags on the 825 RIPE Atlas anchors used as a unicast control [30], confirming the absence of false positives under the c_f bound f or known unicast infrastructure.

On a set of 66 cloud-provider IP ranges (AWS and Google Cloud) [1, 16], the filter flagged only two instances; the remaining 64 ranges are not categorized as “globally announced” in upstream datasets and are geolocatable as single locations, consistent with our non-flag results. For a larger dataset of 143 “globally announced” CDN prefixes (including Cloudflare), the test flagged 83. The 60 unflagged prefixes are predominantly AWS ranges labeled as global in the metadata but which do not exhibit anycast behavior in a geolocation-relevant sense—a known artifact of upstream dataset labeling [19, §6]—or represent replicas clustered within a single region. Since such clustered replicas behave indistinguishably from unicast targets for CBG++ geolocation, their omission from anycast filtering is harmless.

5.4 Coverage-Stratified Analysis: Short-RTT Selection and Multi-Disk Recovery

Prior works identified low-RTT landmarks as the principal signal for city-level accuracy [14, 22]; Darwich et al. [12] demonstrated that a single closest-RTT landmark suffices in many cases. This consensus raises a question: does short-RTT dominance hold across all coverage conditions, and what do landmarks at longer RTTs contribute? We partition landmarks by their measured RTT to the target into three ranges—short $[0, 5)$ ms, mid $[5, 50)$ ms, and long $[50, \infty)$ ms—and refer to range-restricted selection strategies (e.g., short $K = 1$ for the K landmarks with the lowest RTT within the short range).

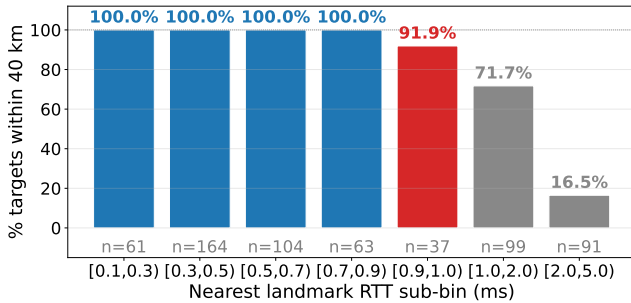


Figure 6: City-level (≤ 40 km) accuracy of short- $K=1$ selection across sub-bins of nearest-landmark RTT in the short range $[0, 5)$ ms. Short- $K=1$ is eligible for $n = 619$ of 795 targets.

In our cross-validation across 795 RIPE Atlas anchor targets, short-RTT selection achieves substantially smaller centroid error than mid-RTT in aggregate (median error 5.79 km versus 279 km, both at $K=1$). This advantage is conditional

on landmark proximity. When a close landmark is available, a single short-RTT landmark is almost always sufficient. For targets with a sub-millisecond nearest landmark ($n=429$), 99.3% achieve city-level accuracy. Despite differences in dataset and landmark composition, our 94.1% rate within 2 ms confirms the 93–95% accuracy reported in prior work [14, 22]. Sub-stratifying within this range (Figure 6), city-level accuracy is deterministic below 0.9 ms in our data (blue, $n=392$, 100%), shows its first failures in the boundary sub-bin $[0.9, 1.0)$ ms (red), and degrades rapidly thereafter—to 71.7% in $[1, 2)$ ms and 16.5% in $[2, 5)$ ms. The 0.9 ms cutoff corresponds to ≈ 90 km of fiber distance (round-trip), consistent with intra-metropolitan landmark-target proximity. Short-RTT selection is eligible for 619 of 795 targets (78%); the remaining 22% have no landmark within $[0, 5)$ ms.

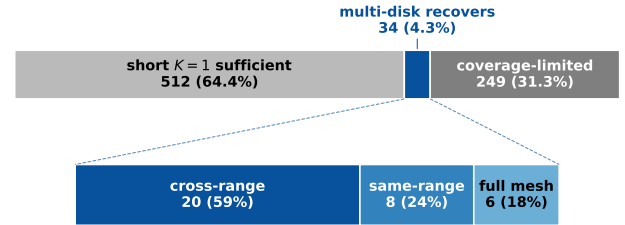


Figure 7: Decomposition of all $n = 795$ targets by city-level (≤ 40 km) achievability into three regimes.

We categorize targets by city-level achievability into three regimes (Figure 7). The dominant regime, comprising 64.4% ($n=512$), reaches city-level accuracy with the closest landmark in the short-RTT range ($K=1$). A further 4.3% ($n=34$) are recovered by multi-disk intersection (geographic distribution of these cases is shown in Figure 12 in Appendix A), of which 59% are driven by cross-range integration—mid-RTT landmarks filling the gap when short-RTT alone is insufficient—while 24% require additional short-RTT landmarks (short $K>1$) and 18% require the full mesh of landmarks. The remaining 31.3% ($n=249$) are coverage-limited within the landmark mesh.

We further decompose these coverage-limited targets into four mechanisms: *spatial sparsity* (52.6%)—no landmark is both geographically close and within short-RTT range; *short-RTT- but-far* (20.1%)—a sub-5 ms landmark exists but is geographically distant; *close-but-slow* (12.9%)—a close landmark exists but its RTT is inflated by last-mile delay; and *intersection-geometry failure* (14.5%)—close-fast landmarks exist but their multi-disk intersection still misses. The first three are not algorithmically recoverable from the landmark mesh: an oracle removal of the offending landmarks in the *short-RTT- but-far* subset ($n=49$) yielded zero city-level recoveries while increasing median centroid error by +175 km,

confirming that these landmarks act as binding constraints even when they do not contain the truth.

The remaining 14.5% (*intersection-geometry failure*) is a distinct, addressable case. These targets have landmarks both close and fast, but the close-fast landmarks happen to lie predominantly in the same direction relative to the target, leaving a wide angular gap on the opposite side. This geometric-dilution-of-precision pattern, analogous to GPS satellite GDOP, motivates an angular- diversity criterion in Stage 3 selection (§4.3).

5.5 Selection: Stage-by-stage Performance (3-stage)

5.5.1 First Stage. For the random baseline, we drew 100 independent seeded VP samples and report the mean across seeds. We verified convergence by tracking the running mean and standard error of each metric as a function of the number of seeds: containment and jurisdictional metrics stabilized within 50 seeds (relative SE < 1%), and all remaining metrics reached relative SE $\leq 3\%$ by 100 seeds. In Table 3, random values are mean $\pm$ SEM across 100 independent seeds; ϵ -net is deterministic per target, and the diversity baselines are deterministic up to negligible tie-break variance.

Table 3: Cross-validation performance of VP-selection methods (on Jan 22, 2026).

Metric	Random	ϵ -net	Div. (AS)	Div. (Country)
<i>Containment / jurisdictional</i>				
In-region (%)	93.6 $\pm$ 0.1	88.6	90.7	90.5
Country match (%)	99.3 $\pm$ 0.02	99.4	98.7	98.6
<i>Accuracy</i>				
Median error (km)	121.6 $\pm$ 2.2	26.6	191.6 $\pm$ 0.9	184.2
Within 25 km (%)	31.8 $\pm$ 0.2	48.7	10.5	9.7
Within 50 km (%)	39.0 $\pm$ 0.2	55.1	15.0	18.2
Within 100 km (%)	47.3 $\pm$ 0.3	61.2	26.1	29.3
<i>Region precision</i>				
Median area (km ²)	268,700 $\pm$ 7,200	17,843	823,897	447,542
Median #countries	2.9 $\pm$ 0.03	1	7	4

To test whether topological or jurisdictional metadata can supplant or complement RTT-space ϵ -net, we evaluated k -center greedy on geodesic distance combined with a one-anchor-per-cluster restriction (ASN or country as the cluster key). At each step, k -center greedy adds the anchor whose nearest already-selected neighbour is farthest—Gonzalez’s farthest-first traversal [?], a 2-approximation to the max-min pairwise-distance objective; the cluster restriction further forces selection from clusters not yet represented, until every cluster has at least one anchor.

The two cluster choices yield similar point accuracy at $n=100$ —median error of 191.6 km for ASN and 184.2 km for country—both roughly 7 $\times$ worse than ϵ -net’s 26.6 km, and

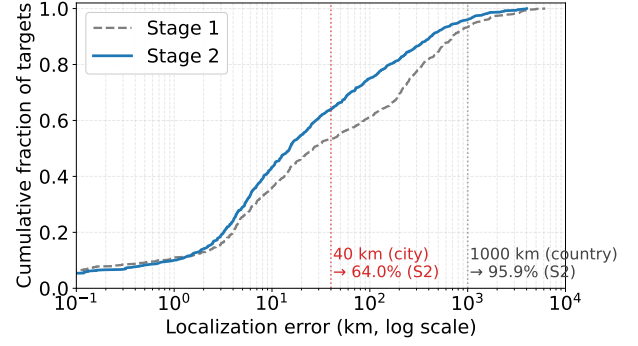


Figure 8: Cumulative distribution of localization error across $N=788$ leave-one-out cross-validation targets.

worse even than uniform random selection at 121.6 km. Cluster choice mildly differentiates regional sharpness (country-restricted intersections span 4 countries vs. 7 and cover roughly half the area), but neither variant approaches ϵ -net, whose intersections are typically confined to a single country with a median area of $\sim 18,000$ km².

The mechanism is geometric: enforcing spread across clusters ensures every selected anchor is far from any target, so every RTT-distance disk is wide and the intersection inflates—the opposite of what CBG benefits from. ϵ -net in RTT space, by contrast, retains low-RTT (and therefore tight-disk) anchors by construction. We conclude that for city-level accuracy on the RIPE anchor mesh, neither ASN nor country metadata improves selection on top of RTT-space ϵ -net.

5.5.2 Second Stage. Figure 8 reports the cumulative distribution of localization error across our cross-validation targets, with the Stage 1 (ϵ -net only) and Stage 2 (RTT-refined) curves overlaid. Stage 2 reaches city-level accuracy—within 40 km of the true location—for 64% of targets, well above the Stage 1 baseline. The improvement concentrates in the mid-range, where Stage 1 places the centroid in the right country but the wrong metropolitan area; the RTT refinement of Stage 2 is precisely what pulls these misses down to city level. The remaining targets sit in the right tail and constitute the cases for which Stage 3 (§4.3) is invoked—those where the anchor mesh alone fails to produce a city-level fix.

5.5.3 Third Stage. Figure 9 shows the deployed pipeline’s effect on the overall localization-error distribution: Stage 3 lifts the fraction of targets reaching city-level accuracy from 64% to 70%, a 6-pp improvement that concentrates in the 40–200 km range—precisely the targets where Stage 2 placed the centroid in the right neighbourhood but missed the city. Figure 10 characterises this conditional behaviour. Stage 3’s gains are highly bucket-dependent: targets that Stage 2 has already constrained to a small, well-located region (the *close-tight* quadrant) are recovered at 43%, while targets with a far

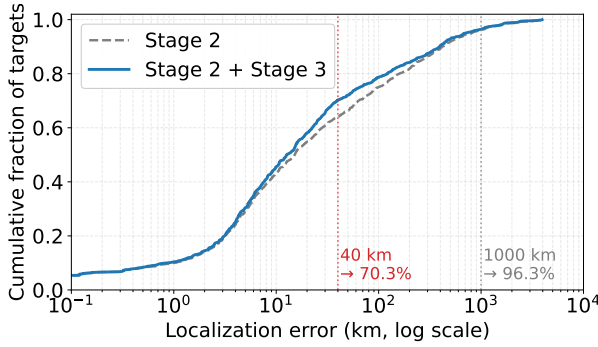


Figure 9: Localization error CDF before vs. after Stage 3 invocation.

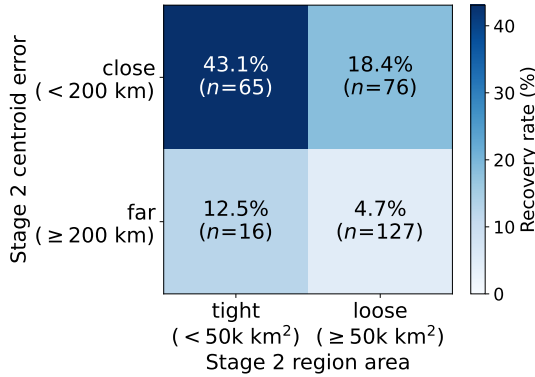


Figure 10: Stage 3 city-level recovery rate, conditioned on Stage 2's baseline geometry. Targets are bucketed by Stage 2 centroid error (close: <200 km, far: ≥200 km) and Stage 2 region area (tight: <50,000 km², loose: ≥50,000 km²).

centroid and a loose region see almost no benefit (5%). The first quadrant captures the ϵ -net's natural limit—a region whose centroid is approximately right and whose shape is amenable to refinement—and is where Stage 3's Atlas disks add useful constraint. Outside it, additional probes either tighten the wrong neighbourhood (close-loose) or sit too far from truth to materially shift the centroid (far-loose).

5.6 Calibration Freshness: Bounding the Cost of Staleness

A longitudinal CBG deployment cannot realistically refresh every anchor daily due to measurement overhead. We therefore investigate the temporal stability of calibration. Specifically, how long a fit remains valid before the resulting LP disks meaningfully degrade geolocation accuracy. For 445 anchors with consistent daily LP fits over an 80-day window (Jan 23 – Apr 14, 2026), we compute the gap between the

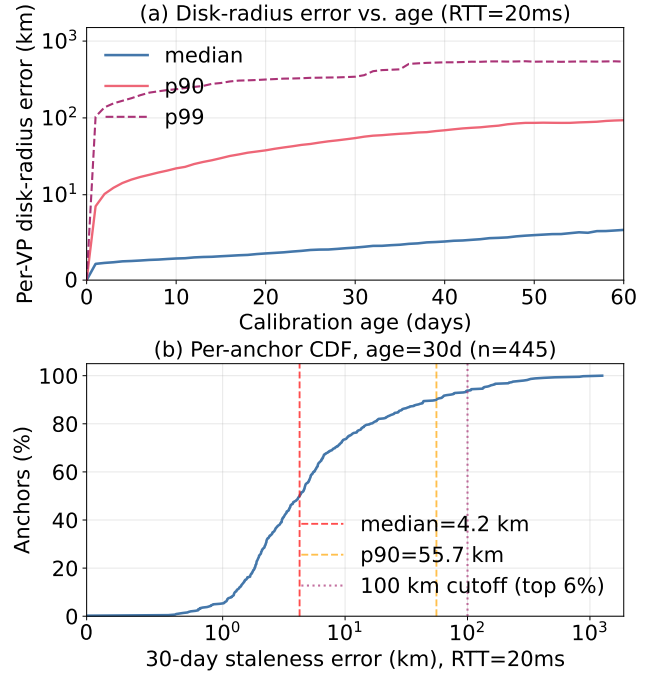


Figure 11: Per-anchor disk-radius error introduced by an age- k calibration over 80 days $\times$ 445 anchors. (a) Median, p90, p99 across all (anchor, D) pairs at RTT=20ms. (b) Per-anchor median 30-day staleness; about 6% exceed the 100 km Tier-3 cutoff.

calibration line fit on day D and day $D + k$, evaluated at a typical RTT (20 ms) and reported as disk-radius error (km).

The error distribution is heavy-tailed (Fig. 11). At a calibration age of $k = 30$ days, the median anchor's disk drifts by 3.8 km, the 90th percentile by 55 km, and a small 99th percentile tail exceeds 100 km within 24 hours. Three operational regimes emerge: $\sim 27\%$ of anchors pass both Augmented Dickey-Fuller (ADF) stationarity and Ljung-Box tests, showing no detectable drift; $\sim 67\%$ drift slowly enough that a 30-day refresh cycle maintains a median error below 5 km; and $\sim 6\%$ accumulate > 100 km of error within 30 days, appearing fundamentally unstable under the current pipeline.

Smoothing is not a fix. Applying a centered MA(5) to the calibration series appears to “recover” 26% of the non-stationary subset to stationarity. However, the same MA(5) reclassifies 27% of *already-stationary* anchors as non-stationary. This symmetry indicates that the apparent recovery is an artifact of the moving average shrinking series variance (by 71%) and introducing artificial autocorrelation, rather than a genuine reduction in underlying drift.

Refresh Policy. These regimes suggest a tiered scheduling strategy. **Tier 1** ($\sim 27\%$) requires only a single cached fit, updated on demand. **Tier 2** ($\sim 67\%$) follows a 30-day refresh

	Full landmarks	Stage 1+2 only	Our pipeline
Anchors used / target	≈ 788	≈ 77	≈ 77
Expected pings / target	≈ 788	≈ 77	≈ 80
Median error	12.6 km	15.2 km	12.7 km
City-level (≤ 40 km)	65.5%	64.0%	70.3%

Table 4: End-to-end comparison ($N=788$ leave-one-out targets).

cadence, consistent with the 45-day median stable segment found via PELT change-point detection on non-stationary anchors. **Tier 3** ($\sim 6\%$) is best excluded from longitudinal use at city-level resolution, since their disk-radius error exceeds 100 km within 30 days. We leave alternative recovery strategies for Tier-3 anchors as future work.

5.7 System: End-to-end performance

Our three-stage pipeline trades exhaustive anchor probing for a small, well-chosen subset—about 77 anchors per target plus on average ≈ 3 additional probe pings on the $\sim 36\%$ of targets where Stage 3 is invoked, for ≈ 80 measurements in total. Against the full-landmark CBG baseline’s ≈ 929 anchor pings, our pipeline still exceeds the baseline at the city-level threshold by 4.8 percentage points (70.3% vs. 65.5%) and matches it at the median (12.7 vs. 12.6 km), as summarised in Table 4.

6 Related Work

6.1 Latency-based IP Geolocation

Latency-based geolocation has been widely studied through both geometric and statistical approaches. Constraint-Based Geolocation (CBG) [18] represents a geometric approach that estimates a target’s location by applying distance constraints derived from RTT measurements to known landmarks. Specifically, CBG constructs disks on a sphere and computes their intersection to localize the target, relying on calibration functions to map RTTs to geographic distances. CBG++ [34] extends this framework by improving both calibration and intersection procedures, leading to more accurate and robust estimates. Octant [35] generalizes the constraint-based formulation by introducing both positive and negative constraints, regions where the target must or cannot reside, and supporting intersections of arbitrary convex and concave regions rather than only disks.

Beyond geometric formulations, prior work has explored alternative modeling paradigms. Spotter [23] adopts a statistical perspective by modeling RTT-to-distance relationships as probability distributions and combining per-landmark likelihoods through probabilistic inference to estimate the most likely location.

6.2 Vantage Point Selection

6.2.1 Short-RTT-based selection. Several approaches aim to identify vantage points geographically close to the target by exploiting latency measurements. RIPE IPmap’s single-radius engine [14] infers a target’s location from the Atlas probe with the lowest RTT, relying on the “lucky” presence of a nearby probe [33]; the IPmap evaluation reports that city-level accuracy drops from 80.6% to 55.8% in large transit ASes where such probes are less likely to exist. TinyG [22] formulates probe selection as a sequential decision problem, choosing probers in successive rounds to maximize expected entropy reduction with the goal of identifying a prober within a small RTT (~ 2 ms) of the target, a threshold shown to correlate with credible city-level accuracy [14]. Replication studies [12] further confirm that even a single well-chosen vantage point suffices when one exists nearby, yielding accuracy comparable to using thousands of probes, and explicitly identify the absence of close vantage points as the dominant failure mode of latency-based geolocation.

However, even with these advances, latency-based selection still faces three open challenges: a sufficiently close probe must exist and be identifiable without prior knowledge of the target; short-RTT measurements can be inflated by queuing, last-mile latency, and routing circuitousness [14]; and inference based on a single nearest probe offers limited means to flag such distortions when they occur. These observations suggest that approaches incorporating signal from multiple probes—including more distant ones—can complement nearest-probe selection, particularly in regions where close probes are scarce.

6.2.2 Diversity-based selection. An alternative line of work focuses on selecting a diverse set of vantage points to improve global coverage and representativeness. Metis [2] proposes a data-driven approach that selects Atlas probes based on topological distance metrics (RTT, AS-path length, or IP hops), improving geographic and RIR coverage compared to Atlas’s default random selection. For instance, increasing country coverage by up to 59% and tripling representation from underrepresented regions. In the geolocation setting, Cho et al. [7] evaluate landmark selection strategies that combine AS-level clustering with geographic-distance maximization, showing that a carefully chosen subset of 36% of available RIPE Atlas anchors can reproduce the geolocation judgments of the full pool.

By design, both approaches optimize for global diversity rather than per-target proximity, and the resulting probe sets offer no inherent guarantee that any selected probe lies near a particular target.

6.2.3 New signal-based selection. GeoResolver [29] uses DNS redirection patterns via ECS queries to identify geographically proximate vantage points, assigning the location of the landmark with the lowest RTT. While this approach significantly reduces probing budget and operates even with limited ECS support, it fundamentally inherits the short-RTT premise. Its accuracy remains contingent on the existence of a vantage point within 2 ms of the target—the same structural constraint that binds the other short-RTT methods discussed above.

6.3 Anycast Detection

Approaches to distinguishing anycast from unicast addresses fall into three families. iGreedy [8], which we adopt (§3.4), detects targets whose multi-vantage latency measurements are physically inconsistent with a single location—a speed-of-light (SoL) violation. MAnycast2 [32] flags targets whose responses arrive at multiple sites of an anycast probing infrastructure. LACeS [19] combines both, using MAnycast2 to identify candidate anycast prefixes and SoL violations to confirm them. All three share the limitation that false negatives arise when replicas are geographically or topologically clustered [8, 19]. These tools were primarily designed for comprehensive anycast censuses, complementing commercial datasets such as IPinfo [21] and BGPtools [3]; we apply the SoL test only as a pre-filter, not for enumeration.

6.4 IPv6 Geolocation

As IPv6 adoption climbs, adapting geolocation frameworks has become a priority. While Darwich et al. [12] deferred IPv6 adaptation to future work due to landmark sparsity, Rimplinger et al. [29] successfully ported point-estimate methods to IPv6 using DNS redirection. We complement these efforts by evaluating the generalizability of our LP-calibrated CBG pipeline via a direct IPv4-vs-IPv6 comparison (§5.1). Recent work also targets IPv6 data scarcity; for instance, Cheng et al. [6] mine landmarks via WiFi-BSSID associations to handle frequent prefix rotation.

7 Limitations

Target representativeness and geographic skew. RIPE Atlas anchors are institutionally hosted infrastructure with well-conditioned routing, atypical of residential or mobile endpoints where last-mile delay and address dynamism likely yield larger localization errors. The mesh is also geographically skewed—our sanitized mesh contains 489 European anchors but only 27 in Africa (§??). Aggregate accuracies reported here should therefore not be interpreted as universal CBG++ performance, particularly in under-served regions.

Coverage-limited algorithmic floor. A floor of approximately 31 % of targets remains coverage-limited under our

anchor mesh, with the dominant cause (52.6 % of these) being spatial sparsity—no landmark sufficiently close to the target (§5.4. This fraction is not reducible by any of the selection or intersection strategies we evaluated, and would require additional probe deployment, particularly in under-served regions. Stage 3 expansion to the broader RIPE Atlas probe pool (§4.3) recovers only a small additional fraction of these targets, structurally limited by the heterogeneity and lower reliability of probe-class measurements relative to anchors.

Methodological caveats. The anycast pre-filter identifies globally-distributed replicas via a lightweight three-pair violation test (§3.4); it cannot detect within-region replicas (e.g., multi-PoP deployments within a single country) and may miss borderline violations masked by last-mile delay. Anchor coordinates are administrative claims verified by RIPE NCC rather than independently measured, so coordinated administrative errors would not be detected by our framework. Finally, while our calibration is designed for periodic re-verification under network change, comprehensive longitudinal validation across multiple mesh snapshots is left to future work.

8 Ethics

We anticipate low ethical risk from this work. Our analysis uses RIPE Atlas’s pairwise anchor-mesh measurements, routinely collected by infrastructure operated by volunteers and made available to researchers. Anchor metadata, including geographic coordinates, are publicly disclosed by RIPE NCC; we do not introduce additional identification risk beyond what is already publicly available. Where supplementary measurements were issued, they operated at low rates comparable to routine network diagnostics, and we are not aware of any service disruption. Targets were limited to RIPE Atlas anchors, and no attempts were made to access private services or identify individual users. Data and code will be released upon publication to support reproducibility.

9 Conclusion

In this paper, we presented a multi-stage landmark selection framework that achieves city-level IP geolocation accuracy while significantly reducing the operational burden of longitudinal monitoring. By integrating a lightweight incremental re-verification mechanism, our approach ensures that high-fidelity re-localization is triggered only upon detecting significant RTT shifts, providing a scalable and cost-effective solution for maintaining accurate IP-to-location mappings over time.

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Appendix

A Geometric Case Studies of Multi-Disk Recovery

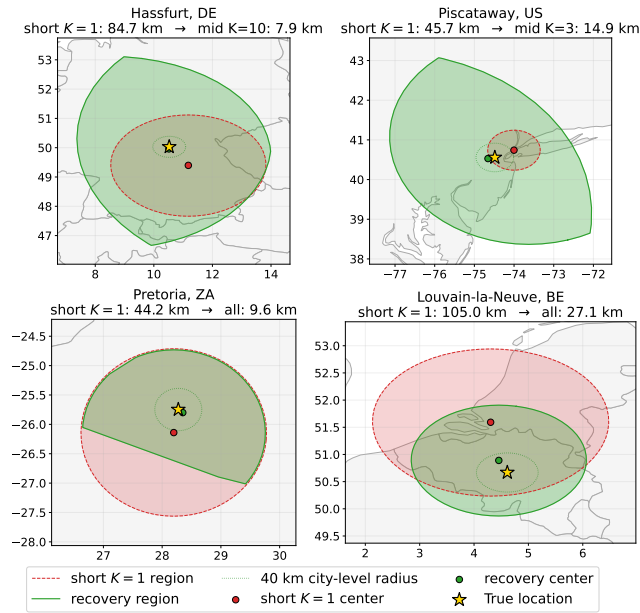


Figure 12: Geometric view of four multi-disk recovery cases. In each panel, the red dashed region is the CBG feasible area produced by short- $K=1$ selection (which fails the 40 km city-level test), and the green solid region is the area produced by the recovery strategy (which contains the true location, marked by the gold star). Per-panel headers report the target, recovery strategy, and the error before/after recovery.

Figure 12 illustrates why multi-disk intersection recovers targets that short- $K=1$ alone cannot. We pick four representative cases spanning the recovery mechanisms identified in §5.4. Hassfurt (DE) and Piscataway (US) are recovered through cross-range integration with mid-RTT landmarks ($K=10$ and $K=3$, respectively), while Pretoria (ZA) and Louvain-la-Neuve (BE) require the full landmark mesh.

In every panel, the short- $K=1$ centroid is offset from the true location—a single disk imposes only one geometric constraint, so even if it is tight, the centroid drifts whenever the landmark is not perfectly aligned with the target.

In some cases, adding more landmarks (same-range or cross-range) shrinks the feasible region through additional intersections, pulling its centroid onto the true location. The Louvain-la-Neuve (BE) case is the most striking. The short- $K=1$ centroid lies in the Netherlands (105.0 km error, country mismatch), but with the full landmark set the centroid is correctly placed in Belgium (27.1 km error).

The cases also show that recovery works in two distinct ways. Mid-RTT recovery (Hassfurt, Piscataway) discards the short- $K=1$ disk and intersects a different set of (larger) mid-RTT disks; the recovered region can be larger than the original short- $K=1$ disk but is better-centered on the true location. Full-mesh recovery (Pretoria, Louvain-la-Neuve) keeps the short- $K=1$ disk and intersects it with all remaining constraints, producing a tighter, better-centered region.